

Advanced (Habanero)

Q1. Factor the expression $x^3 + 8$

- (A) $(x + 2)(x^2 - 2x + 4)$
- (B) $(x - 2)(x^2 + 2x + 4)$
- (C) $(x + 2)(x^2 + 2x + 4)$
- (D) $(x - 2)(x^2 - 2x + 4)$
- (E) None of the above

Q2. Which of the following is a factorization of $27x^3 - 8y^3$

- (A) $(3x - 2y)(9x^2 + 6xy + 4y^2)$
- (B) $(3x + 2y)(9x^2 - 6xy + 4y^2)$
- (C) $(3x - 2y)(9x^2 - 6xy + 4y^2)$
- (D) $(3x + 2y)(9x^2 + 6xy + 4y^2)$
- (E) None of the above

Q3. Factor $64 - x^3$

- (A) $(4 - x)(16 + 4x + x^2)$
- (B) $(4 + x)(16 - 4x + x^2)$
- (C) $(4 - x)(16 - 4x + x^2)$
- (D) $(4 + x)(16 + 4x + x^2)$
- (E) None of the above

Q4. Which expression is equivalent to $(x + 2)(x^2 - 2x + 4)$

- (A) $x^3 - 8$
- (B) $x^3 - 4$
- (C) $x^3 + 8$
- (D) $x^3 - 2$
- (E) None of the above

Q5. Factor $x^3 + 27$

- (A) $(x + 3)(x^2 - 3x + 9)$
- (B) $(x - 3)(x^2 + 3x + 9)$
- (C) $(x + 3)(x^2 + 3x + 9)$
- (D) $(x - 3)(x^2 - 3x + 9)$
- (E) None of the above

Q6. Factor $8x^3 - 1$

- (A) $(2x - 1)(4x^2 + 2x + 1)$
- (B) $(2x + 1)(4x^2 - 2x + 1)$
- (C) $(2x - 1)(4x^2 - 2x + 1)$
- (D) $(2x + 1)(4x^2 + 2x + 1)$
- (E) None of the above

Q7. Which of the following is a factorization of $125x^3 - 64y^3$

- (A) $(5x - 4y)(25x^2 + 20xy + 16y^2)$
- (B) $(5x + 4y)(25x^2 - 20xy + 16y^2)$
- (C) $(5x - 4y)(25x^2 - 20xy + 16y^2)$
- (D) $(5x + 4y)(25x^2 + 20xy + 16y^2)$
- (E) None of the above

Q8. Factor $x^3 - 64$

- (A) $(x - 4)(x^2 + 4x + 16)$
- (B) $(x + 4)(x^2 - 4x + 16)$
- (C) $(x - 4)(x^2 - 4x + 16)$
- (D) $(x + 4)(x^2 + 4x + 16)$
- (E) None of the above

Open Ended Questions (FRQ)

Q9. Factor $27x^3 + 8y^3$

Q10. Factor $x^3 - 125$

Chef's Tips

- Remind students to check their work by multiplying the factors.
- Encourage students to look for common factors before factoring.
- Suggest that students use the sum and difference of cubes formulas to factor expressions.